Introduction

Due to magnetic field inhomogeneities EPI can suffer from geometric distortions and signal dropouts. The problem of geometric distortions can be adressed by acquiring field maps during the fMRI session and using SPM's FieldMap Toolbox for undistorting the EPI images (Hutton et al, 2002). The toolbox can be used either interactively or scripted.

Acquisition of field maps

Ideally a field map should be acquired before or after each fMRI scan, however the first field map has to be acquired not before the first fMRI scan. If you are using a field map for different fMRI runs make sure that the field map is aligned properly with the EPI images.

Sequence: For acquiring a field map use the sequence 'gre_field_mapping_1acq_rl' from the routine sequence folder 'User\FIL\Routine'. Select the respective version from there depending on the used coil. The acquisition takes approximately 2.5 minutes and creates two series of images.

Setup of the field map:

- The number of slices should remain 64 and the orientation transverse.
- The field-of-view should cover the whole head.
- Copy the shim setting used for the EPI acquisition: Click with the right mouse button on the EPI sequence that was acquired previously, select 'Copy Parameter', and then select 'Adjust Volume'. A warning message pops up asking you whether you wish a manual or automatic adjustment. Select 'Manual Adjustment'.

Using the FieldMap toolbox for undistorting EPI images

You can use the FieldMap toolbox via the SPM user interface either by using the **Batch Editor** or by using the **GUI**. Or you can you can use the FieldMap toolbox by **Scripts**.

For the following it is quite helpful to have some information off pat.

For the field map reconstruction: Short TE, Long TE and if the field map is EPI-based. These parameters can either be found in the protocol of the used field map sequence or can be looked up on the wiki page of the respective sequence. At the FIL (Trio, Quattro and Penta) the parameters are usually TE1=10ms, TE2=12.46ms and EPI-based=no.

For the EPI unwarping: Polarity of phase-enode blips, if you like to apply Jacobian modulation and the total EPI readout time. These parameters can be found at the respective EPI sequence descriptions of the wiki.

Using the FieldMap toolbox via Batch Editor

Start the Batch Editor via selecting 'Batch' on the SPM user interface. Then select via the menue of the Batch Editor 'SPM' > 'Tools' > 'FieldMap' > 'Presubtracted Phase and Magnitude Data'.

1. Select 'New: Subject'
2. For the created subject select 'Phase Image'. Note: The Phase Image (phase difference image) is the second image of the second series of the field map acquisition.
3. Select 'Magnitude Image'. Note: The Magnitude Images can be found in the first series of the field map acquisition. Take the first one.
4. For the 'FieldMap Defaults' one can choose between 'Defaults values' and 'Defaults File'. In case of 'Defaults values' you have to enter the fieldmap information from above manually and can change some additional options. In case of 'Defaults File' you have to choose a suitable default file, e.g. pm_defaults_primsa - the correct file is listed on the wiki page for each sequence - note there are different pages and pm_defaults files for the Trio and Prisma systems!
5. 'EPI Sessions': Select the first volume of each EPI session that will be ultimated realigned and unwarped using the selected field map data. If you only collected a single field map for all sessions, you must select a new EPI for each session so that a voxel displacement map can be generated for each session.
6. Select 'Match VDM to EPI' to make sure the field map is matched to the EPI time series. This will update the positional information for the voxel displacement map.
7. 'VDM filename extension' etc.
8. Select 'Write unwarped EPI' if you wish to have an unwarped EPI written at this stage (possibly for assessment). This will be overwritten when running Realign & Unwarp on the whole time series.
9. Select 'Select anatomical image for comparison' if you would like a visual assessment of the unwarped EPI with anatomy. This is not a necessary part of the processing and can be left empty.
10. If you have selected an anatomical file in the previous step choose for 'Match anatomical image to EPI' 'match anat' otherwise 'none'.

Using the FieldMap toolbox via the GUI

Start the FieldMap toolbox by clicking on >Toolboxis>FieldMap

1. On the pull-down menu on the top right you can select an appropriate defaults file. The default files should contain the correct values for the default fieldmap and EPI sequences on each scanner, e.g. pm_defaults_primsa - the correct file is listed on the wiki page for each sequence - note there are different pages and pm_defaults files for the Trio and Prisma systems! These will not be correct if using other EPI sequences.
2. If you need to change the parameters, you can do this via the GUI. You can also create your own customised version of a pm_defaults file containing the required values. To do this, copy one of the pm_defaults_*.m files to a new file called (for example) pm_defaults_mydefaults.m and edit the values that need changing.
3. Select the top 'Load Phase' button and select the corresponding phase image. Note: The phase image (phase difference image) is the second image of the second series of the field map acquisition. When asked if you want to scale to radians, respond Yes. This will write out a file with the prefix sc which is the phase in radians and use this file for further processing.
4. Select the top 'Load Mag.' button and select the first of the two magnitude files. Note: The Magnitude Images can be found in the first series of the field map acquisition.
5. You can ignore the second pair of buttons with 'Load Phase' and 'Load Mag.' as these aren't required for the field maps acquired at the FIL.

6. The 'Mask brain' option extracts the brain from the magnitude image which can then be used as a mask for the field map. To do this, select Yes. It takes a little longer to process, but results in a much tidier field map.
7. Press 'Calculate'. This will display a field map in an SPM graphics window which should be bright to dark from anterior to posterior around sinus, dark at front of temporal lobes going bright towards back of head. If you click in this image, you will see the corresponding field value in Hz in the control panel.
8. You can save the field map (in Hz) by selecting the 'Write' button. This will write out an fpm5_*.img file which can be loaded into FieldMap at a later date if necessary.
9. A vdm5_*.img file is written to the same directory as the field map data. This is the voxel shift map which you can use in the Realign & Unwarp step to apply the distortion correction.
10. It is important that the field map data is aligned to the first (non-dummy) EPI volume of each EPI time series.
11. Using the 'Load EPI image' button at the buttom of the GUI, select the first EPI in the fMRI sequence that will be 'Realigned and Unwarped'. This will display the original and the distortion corrected EPI in the SPM graphics window.
12. Select Match 'VDM to EPI' to make sure the field map is matched to the EPI time series. This will update the positional information for the vdm5_* file.

Some Literature

Jezzard P and Balaban RS (1995) Correction for geometric distortions in echoplanar images from B0 field variations. Magn Reson Med 34:65-73

Andersson JLR, Hutton C, Ashburner J, Turner R, Friston K (2001) Modelling geometric deformations in EPI time series. NeuroImage 13:903-919

Hutton C, Bork A, Josephs O, Deichmann R, Ashburner J, Turner R. (2002). Image distortion correction in fMRI: A quantitative evaluation. NeuroImage 16:217-240

Jenkinson M. (2003). Fast, automated, N-dimensional phase-unwrapping algorithm. MRM 49:193-197

Hutton, C., Deichmann, R., Turner R., Andersson, J. L. R., (2004). Combined correction for geometric distortion and its interaction with head motion in fMRI. Proceedings of ISMRM 12, Kyoto, Japan